2.3 Commutative Rings; the Integers Are One

Sections 2.1 and 2.2 built $\mathbb{Z}$ with well-defined operations $+$, $\cdot$ and $-$. Why was it built at all? $\mathbb{N}$ already adds and multiplies by all the usual laws but one: it has no negatives, since $1+x = x+1 \neq 0$ for every $x \in \mathbb{N}$ (Theorem 1.2.3 (b), Corollary 1.2.2). This section collects the laws of school algebra into one definition, a **commutative ring**, checks them for $\mathbb{Z}$ by reducing each to a law of $\mathbb{N}$, and derives the consequences that hold in every commutative ring.

2.3.1 The laws of a ring

Expanding brackets or moving a term across an equation uses a short list of laws: sums and products may be reordered and regrouped, 0 and 1 do nothing, every element has a negative, and products distribute over sums. The list holds in $\mathbb{Z}$, it will hold in $\mathbb{Q}$, and a result proved from the list alone holds wherever the list does. It deserves a name.

A commutative ring is a set where you may add, subtract and multiply by the usual rules; a field is one where you may also divide by anything non-zero.

Definition 2.3.1. Commutative Ring, Field

A **commutative ring** is a set R with two functions, **addition** $+\colon R \times R \to R$ and **multiplication** $\cdot\colon R \times R \to R$ (written $a \cdot b = ab$), such that clauses (F1)–(F7) and (F9) of the following list hold; clause (F8) is used only for fields.

(F1) $a + b = b + a$ **for all** $a, b \in R$.
(F2) $(a + b) + c = a + (b + c)$ **for all** $a, b, c \in R$.
(F3) **There is** an element $0 \in R$ with $a + 0 = a$ **for every** $a \in R$.
(F4) **For every** $a \in R$ there is an element $-a \in R$ with $a + (-a) = 0$.
(F5) $(ab)c = a(bc)$ **for all** $a, b, c \in R$.
(F6) $ab = ba$ **for all** $a, b \in R$.
(F7) **There is** an element $1 \in R$ with $a \cdot 1 = a$ **for every** $a \in R$.
(F8) **For every non-zero** $a \in R$ there is an element $a^{-1} \in R$ with $aa^{-1} = 1$.
(F9) $a(b + c) = ab + ac$ **for all** $a, b, c \in R$.

A **field** is a commutative ring in which, **in addition**, $1 \neq 0$ and (F8) holds. We write $a - b \coloneqq a + (-b)$, and in a field $a/b \coloneqq ab^{-1}$ for $b \neq 0$.

In words: (F1)–(F4) say that addition is commutative and associative, with a neutral element 0 and negatives; (F5)–(F7) say the first three things about multiplication; (F9) links the two and lets us expand brackets. (F8) asks for inverses of **non-zero** elements only, and only in a field. The notations $-a$ and a^{-1} anticipate that these elements are unique, which we prove below. These are the axioms (F1)–(F9) of the linear-algebra companion, Chapter 0, and of the ordered fields of *Analysis on the Real Line*, Chapter 1, in the same numbering, except that there (F7) includes $1 \neq 0$.

Examples.

- $\mathbb{Z}$, with the operations of Definition 2.2.1: the next theorem.

- **The zero ring** $R := \{0\}$, with $0 + 0 := 0$, $0 \cdot 0 := 0$ and $1 := 0$. Every axiom other than (F8) is an equation in R, and every such equation reads $0 = 0$. So R is a commutative ring, with $1 = 0$, and not a field. This degenerate ring is why several results below assume $1 \neq 0$.

A non-example by minimal change. The set $\mathbb{N}$, with its addition and multiplication from Chapter 1, satisfies (F1)–(F3), (F5)–(F7) and (F9), by Theorem 1.2.3 and Definition 1.2.1 (a). That one missing clause, (F4), is why $\mathbb{Z}$ had to be built.

Why these clauses. No axiom says that $ab = 0$ forces $a = 0$ or $b = 0$: that is true in $\mathbb{Z}$ but fails in other commutative rings (the warning below), so it must be proved where it holds. The condition $1 \neq 0$ is kept out of the ring axioms, so that the zero ring is a ring, and put into the field axioms, so that $\{0\}$, where (F8) holds vacuously, is not a field.

Warning. A commutative ring may have **zero divisors**: non-zero a, b with $ab = 0$. The integers modulo 6 of *Sets and Functions*, Section 5.2 (built there from the school integers, with well-defined operations by Sets and Functions, Proposition 5.2.5) form a commutative ring: each of (F1)–(F7) and (F9) passes from the school integers to classes through representatives, a check we do not need and do not carry out here. In this ring $[2] \neq [0]$ and $[3] \neq [0]$, yet $[2][3] = [6] = [0]$. So "$ab = 0$, hence $a = 0$ or $b = 0$" is not a consequence of the ring axioms (F1)–(F7) and (F9). In a field it is, since $b = a^{-1}(ab)$ when $a \neq 0$ (Section 4.1). In an **ordered** ring it is a theorem, proved in Section 2.4 from the order axioms.

2.3.2 The integers form a commutative ring

To prove a law for classes, compute both sides through representatives; the two pairs of natural numbers are then equal or related, and the classes agree.

Theorem 2.3.1. The Integers Form a Commutative Ring

$\mathbb{Z}$, with the addition, multiplication and negation of Definition 2.2.1, is a commutative ring, with zero $[(0,0)]$ and one $[(1,0)]$, and in it $1 \neq 0$.

Idea (Scratch work). Take (F9), the law with the most terms.

Givens	Goal
$x, y, z \in \mathbb{Z}$	$x(y+z) = xy + xz$
$x = [(a,b)]$, $y = [(c,d)]$, $z = [(e,f)]$: every class has a representative (Sets and Functions, Definition 5.2.4)	two classes, each computed by Definition 2.2.1 from these representatives
left: $(a(c+e)+b(d+f),\ a(d+f)+b(c+e))$; right: $((ac+bd)+(ae+bf),\ (ad+bc)+(af+be))$	equal pairs, or related pairs
distribute in $\mathbb{N}$, then regroup four terms	the same pair: the classes are equal

The operations are functions (Proposition 2.2.2), so any representatives may be used. Only (F4) will need $\sim$; only (F5) is long.

Proof. By Proposition 2.2.2, $+$ and $\cdot$ are functions $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$. Let $x, y, z \in \mathbb{Z}$; by Sets and Functions, Definition 5.2.4, $x = [(a, b)]$, $y = [(c, d)]$, $z = [(e, f)]$ with $a, \dots, f \in \mathbb{N}$. By Definition 2.2.1, each side of each axiom is the class of the pair shown in the table. "Equal" means the same pair of natural numbers, by Theorem 1.2.3 and Definition 1.2.1 (a), (b). "Related" means related by $\sim$ (Definition 2.1.1: $(p, q) \sim (r, s)$ means $p + s = q + r$), so the classes agree by Sets and Functions, Theorem 5.2.1 (b).

Axiom	left side, as a pair	right side, as a pair	why the classes agree
(F1) $x + y = y + x$	$(a + c, b + d)$	$(c + a, d + b)$	equal: $+$ is commutative on $\mathbb{N}$
(F2) $(x + y) + z = x + (y + z)$	$((a+c)+e, (b+d)+f)$	$(a+(c+e), b+(d+f))$	equal: $+$ is associative on $\mathbb{N}$
(F3) $x + [(0, 0)] = x$	$(a + 0, b + 0)$	(a, b)	equal: $m + 0 = m$
(F4) $x + (-x) = [(0, 0)]$	$(a + b, b + a)$	$(0, 0)$	related: $(a+b)+0 = (b+a)+0$
(F5) $(xy)z = x(yz)$	see below	see below	equal, after expanding (below)
(F6) $xy = yx$	$(ac + bd, ad + bc)$	$(ca + db, cb + da)$	equal: $\cdot$ and $+$ are commutative
(F7) $x \cdot [(1, 0)] = x$	$(a \cdot 1 + b \cdot 0, a \cdot 0 + b \cdot 1)$	(a, b)	equal: $n \cdot 1 = n$, $n \cdot 0 = 0$, $n + 0 = 0 + n = n$
(F9) $x(y + z) = xy + xz$	$(a(c + e) + b(d + f), a(d+f)+b(c+e))$	$((ac + bd) + (ae + bf), (ad + bc) + (af + be))$	equal: distribute, then reorder the four terms

For (F5), write $xy = [(g, h)]$ with $g \coloneqq ac+bd$, $h \coloneqq ad+bc$. Then $(xy)z = [(ge+hf, gf+he)]$, and in $\mathbb{N}$

$$ge + hf = ace + bde + adf + bcf,$$
$$gf + he = acf + bdf + ade + bce.$$

On the other side, $yz = [(ce+df, cf+de)]$, and $x(yz)$ is the class of the pair whose coordinates are

$$a(ce + df) + b(cf + de) = ace + adf + bcf + bde,$$
$$a(cf + de) + b(ce + df) = acf + ade + bce + bdf.$$

Both first coordinates are sums of ace, adf, bcf, bde, and both second coordinates of acf, ade, bce, bdf, so the pairs are equal (Theorem 1.2.3 (b)–(g)).

Finally, $1 \neq 0$: if $[(1, 0)] = [(0, 0)]$, then $1 = 0$ by Proposition 2.1.2 (b). But $0 < 1$ in $\mathbb{N}$ (Theorem 1.3.2 (d)), and $<$ is irreflexive (Theorem 1.3.2 (a)), so $1 \neq 0$. This proves the theorem.

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Seven of the eight rows end in "equal". Only (F4) needs $\sim$: the pair $(a + b, b + a)$ records "$(a + b) - (b + a)$", which is zero, without performing the subtraction, and that is the whole point of passing to classes.

The proof used one move eight times. It has a name.

Example. Method: reduce to the level below

Template. To prove a law about classes whose operations are defined on representatives: (1) write each element as a class $[p]$; (2) compute each side by the defining formulas, with any representatives, since the operations are well defined; (3) compare the two representatives at the level below, by its laws: equal, or related; (4) conclude that the classes are equal (Sets and Functions, Theorem 5.2.1 (b)).

When to use it. Laws in $\mathbb{Z}$, with $\mathbb{N}$ below; any structure whose operations are formulas in a known ring (Exercise C1 below).

The usual failure. Claiming equal pairs where only $\sim$ holds (the row (F4)); computing with representatives before the operation is known to be a function; using a law of $\mathbb{Z}$ inside the computation in $\mathbb{N}$, which is circular.

Remark. The zero and one of this ring are $[(0,0)] = \iota(0)$ and $[(1,0)] = \iota(1)$ (Proposition 2.2.3), so after the identification of Section 2.2 they are the elements 0 and 1 of $\mathbb{N}$. With $a - b \coloneqq a + (-b)$, the formula $[(a,b)] = \iota(a) + (-\iota(b))$ of Proposition 2.2.3 reads $[(a,b)] = a - b$ for $a, b \in \mathbb{N}$: the subtraction the pair recorded is now performed in $\mathbb{Z}$.

Quick check. Which class is the negative of $[(2,7)]$? Show by reducing to $\mathbb{N}$ that no other class $[(c,d)]$ satisfies $[(2,7)] + [(c,d)] = [(0,0)]$.

Solution. $-[(2,7)] = [(7,2)]$ (Definition 2.2.1). Now suppose $[(2+c,\ 7+d)] = [(0,0)]$. By Sets and Functions, Theorem 5.2.1 (b) and Definition 2.1.1, $(2+c)+0 = (7+d)+0$, so $2+c = 7+d = 2+(5+d)$. Hence $c+2 = (5+d)+2$ by Theorem 1.2.3 (b), and $c = 5+d$ by cancellation (Proposition 1.3.1 (b)). Then $c+2 = (5+d)+2 = d+7$, that is, $(c,d) \sim (7,2)$, and $[(c,d)] = [(7,2)]$ (Sets and Functions, Theorem 5.2.1 (b)). Proposition 2.3.2 (a) below proves the same uniqueness in every commutative ring, from (F1)–(F7) and (F9) alone.

What the definition does not list must be derived, and the derivations work in every commutative ring at once.

Proposition 2.3.2. Algebra in a Commutative Ring

Let R be a commutative ring, and let $a, b \in R$.

(a) The elements 0 and 1 of (F3) and (F7) are unique, the element $-a$ of (F4) is unique, and $-(-a) = a$.

(b) $a \cdot 0 = 0$.

(c) $(-a)b = a(-b) = -(ab)$ and $(-a)(-b) = ab$.

(d) $(a+b)^2 = a^2 + 2ab + b^2$, where $x^2 \coloneqq xx$ and $2 \coloneqq 1+1$.

Idea. To show that something equals 0, or $-(ab)$, show that it satisfies the **defining equation** of that element and let uniqueness, part (a), finish.

Proof.

(a) Suppose 0 and $0'$ both satisfy (F3). Then $0' = 0' + 0 = 0 + 0' = 0$, by (F3) for 0, (F1), and (F3) for $0'$. Likewise, if 1 and $1'$ both satisfy (F7), then $1' = 1' \cdot 1 = 1 \cdot 1' = 1$, by (F7) and (F6). Next, suppose $a + b = 0$ and $a + c = 0$. Then

$$\begin{aligned} b &= b + 0 = b + (a + c) = (b + a) + c \\ &= (a + b) + c = 0 + c = c + 0 = c, \end{aligned}$$

by (F3), the second hypothesis, (F2), (F1), the first hypothesis, (F1) and (F3). So $-a$ is the **only** element whose sum with a is 0. Finally, $(-a) + a = a + (-a) = 0$ by (F1) and (F4), so a satisfies the defining equation of $-(-a)$, and uniqueness gives $-(-a) = a$.

(b) Let $z \coloneqq a \cdot 0$. By (F3) and (F9), $z = a(0 + 0) = a \cdot 0 + a \cdot 0 = z + z$. Adding $-z$ to both sides,

$$0 = z + (-z) = (z + z) + (-z) = z + (z + (-z)) = z + 0 = z,$$

by (F4), the equation $z = z + z$, (F2), (F4) and (F3). So $a \cdot 0 = 0$.

(c) By (F6) and (F9), $ab + (-a)b = b(a + (-a)) = b \cdot 0 = 0$, the last step by (b). So $(-a)b$ satisfies the defining equation of $-(ab)$, and (a) gives $(-a)b = -(ab)$. By (F6) also $a(-b) = (-b)a = -(ba) = -(ab)$. Hence $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$, using the first identity, the second, and (a).

(d) By (F9) and (F6), $(a + b)(a + b) = (a + b)a + (a + b)b = a^2 + ba + ab + b^2$, and $ba + ab = ab \cdot 1 + ab \cdot 1 = ab(1 + 1) = 2ab$ by (F6), (F7) and (F9). Regrouping by (F1) and (F2) gives $(a + b)^2 = a^2 + 2ab + b^2$.

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Every law of Proposition 2.3.2 could also be checked in $\mathbb{Z}$ by the method of Example (Method: reduce to the level below). The next example puts the two routes side by side.

Example. One law, two routes

Prove that $(-x)y = -(xy)$ for all $x, y \in \mathbb{Z}$ in two ways: (a) by reducing to the level below; (b) from (F1)–(F7) and (F9) alone.

Solution.

(a) Let $x = [(a, b)]$ and $y = [(c, d)]$. By Definition 2.2.1, $-x = [(b, a)]$, so

$$(-x)y = [(bc + ad,\ bd + ac)];$$

and $xy = [(ac + bd,\ ad + bc)]$, so $-(xy) = [(ad + bc,\ ac + bd)]$. The two pairs are equal, since $bc + ad = ad + bc$ and $bd + ac = ac + bd$ by commutativity of $+$ in $\mathbb{N}$ (Theorem 1.2.3

(b)). So the classes are equal.

(b) $\mathbb{Z}$ is a commutative ring (Theorem 2.3.1), so Proposition 2.3.2 (c), with $a = x$ and $b = y$, gives $(-x)y = -(xy)$.

Route (a) needs nothing beyond the definitions of the operations, but it proves the law only for $\mathbb{Z}$. Route (b) proves it in every commutative ring at once, which is why Proposition 2.3.2 is stated for rings.

The argument of Proposition 2.3.2 (a), with multiplication in place of addition, shows that in a **field** a non-zero a has only one inverse: if $ab = 1$ and $ac = 1$, then $b = b \cdot 1 = b(ac) = (ba)c = (ab)c = 1 \cdot c = c$, by (F7), (F5) and (F6). So a^{-1}, and with it a/b, names a single element. And Proposition 2.3.2 (b) explains the zero ring: if $1 = 0$ in a commutative ring, then $a = a \cdot 1 = a \cdot 0 = 0$ for every a, so the ring is $\{0\}$.

Quick check. In a field, let $a \neq 0$. Show that $(a^{-1})^{-1} = a$.

Solution. First, $a^{-1} \neq 0$: otherwise $1 = aa^{-1} = a \cdot 0 = 0$, by (F8) and Proposition 2.3.2 (b), against $1 \neq 0$ in a field. So a^{-1} has an inverse by (F8). And $a^{-1}a = aa^{-1} = 1$ by (F6), so a is an inverse of a^{-1}. By the uniqueness of inverses just proved, it is the inverse: $(a^{-1})^{-1} = a$.

2.3.3 Exercises

2.3.3.1 A. Check your understanding

Exercise. A1

(a) State (F4) with its quantifiers, and say why the zero ring satisfies it.

(b) Name the move by which Theorem 2.3.1 was proved, and say which row of its table needed $\sim$ rather than equality of pairs.

(c) True or false, with a reason: in every commutative ring, $ab = 0$ implies $a = 0$ or $b = 0$.

(d) Prove from (F1)–(F7), (F9) and Proposition 2.3.2 that $(a+b)(a-b) = a^2 - b^2$ in every commutative ring, where $x^2 := xx$.

Solution.

(a) For every $a \in R$ there is an element $-a \in R$ with $a + (-a) = 0$. In $\{0\}$ the only element is 0, and $-0 := 0$ works, since $0 + 0 = 0$.

(b) **Reduce to the level below** (Example (Method: reduce to the level below)). Only (F4) needed $\sim$: the pair $(a + b, b + a)$ is related to $(0, 0)$, but it is not that pair.

(c) False. Give $\mathbb{Z} \times \mathbb{Z}$ the componentwise operations $(x, y) + (u, v) := (x + u,\ y + v)$ and $(x, y)(u, v) := (xu,\ yv)$. It is a commutative ring, with zero $(0, 0)$, one $(1, 1)$ and $-(x, y) = (-x, -y)$: each of (F1)–(F7) and (F9) holds in each coordinate by Theorem 2.3.1. In it $(1, 0)(0, 1) = (1 \cdot 0,\ 0 \cdot 1) = (0, 0)$ by Proposition 2.3.2 (b) and (F6), while $(1, 0) \neq (0, 0)$ and $(0, 1) \neq (0, 0)$, since $1 \neq 0$ in $\mathbb{Z}$. (The integers modulo 6 of the warning after Definition 2.3.1, with $[2][3] = [0]$, illustrate the same, with their

ring laws taken for granted.)

(d) By (F9), and then (F6) and (F9) on each summand ($(a+b)a = a(a+b)$, $(a+b)(-b) = (-b)(a+b)$),

$$\begin{aligned}(a+b)(a+(-b)) &= (a+b)a + (a+b)(-b)\\ &= (aa+ab) + ((-b)a + (-b)b).\end{aligned}$$

By Proposition 2.3.2 (c), $(-b)a = -(ba) = -(ab)$, the last step by (F6), and $(-b)b = -(bb)$. Regrouping by (F1) and (F2), the sum is

$$a^2 + (ab + (-(ab))) + (-(b^2)) = a^2 + 0 + (-(b^2)) = a^2 - b^2,$$

by (F4) and (F3), and the definition $u - v \coloneqq u + (-v)$.

2.3.3.2 B. Practice

Exercise. B1: Three rows in detail

Let $x = [(a,b)]$, $y = [(c,d)]$, $z = [(e,f)]$ be integers. Verify in detail, naming the law of $\mathbb{N}$ used at each step, the rows (F9), (F4) and (F6) of the table in the proof of Theorem 2.3.1.

Solution. All operations are those of Definition 2.2.1, and "Law" refers to Theorem 1.2.3.

(F9). Since $y+z = [(c+e, d+f)]$, $x(y+z) = [(a(c+e)+b(d+f),\ a(d+f)+b(c+e))]$, whose coordinates are $ac+ae+bd+bf$ and $ad+af+bc+be$ by Law (d). And $xy+xz = [((ac+bd)+(ae+bf),\ (ad+bc)+(af+be))]$, whose coordinates are the same sums by Laws (b), (c). So both sides are the class of one pair.

(F4). $x+(-x) = [(a+b, b+a)]$, and $(a+b, b+a) \sim (0,0)$ means $(a+b)+0 = (b+a)+0$ (Definition 2.1.1), true by Definition 1.2.1 (a) and Law (b). So $x+(-x) = [(0,0)]$ by Sets and Functions, Theorem 5.2.1 (b).

(F6). $yx = [(ca+db, cb+da)]$. By Law (f), $ca = ac$, $db = bd$, $cb = bc$, $da = ad$, and by Law (b), $bc+ad = ad+bc$; so the pair is $(ac+bd, ad+bc)$, the pair of xy.

Exercise. B2: Prove or disprove

Prove or disprove.

(a) For all $x, y \in \mathbb{Z}$, $-(x+y) = (-x)+(-y)$.

(b) In every commutative ring, $a \cdot a = a$ implies $a = 0$ or $a = 1$.

(c) The set $\mathbb{N}\times\mathbb{N}$ itself, with $(a,b)+(c,d) \coloneqq (a+c,\ b+d)$ and $(a,b)\cdot(c,d) \coloneqq (ac+bd,\ ad+bc)$, is a commutative ring.

(d) In every commutative ring, $a+b = a$ implies $b = 0$.

Solution.

(a) **True**, by reducing to the level below. Let $x = [(a,b)]$ and $y = [(c,d)]$. By Definition 2.2.1, $-(x+y) = -[(a+c,\ b+d)] = [(b+d,\ a+c)]$, and $(-x)+(-y) = [(b,a)]+[(d,c)] = [(b+d,\ a+c)]$. The pairs are equal, so the classes are.

(b) **False.** In the commutative ring $\mathbb{Z} \times \mathbb{Z}$ with componentwise operations (Exercise A1 (c)), $(1,0)(1,0) = (1 \cdot 1,\ 0 \cdot 0) = (1,0)$ by (F7) and Proposition 2.3.2 (b); and $(1,0)$ is neither the zero $(0,0)$ nor the one $(1,1)$, since $1 \neq 0$ in $\mathbb{Z}$. (In the integers modulo 6, ring laws taken for granted as in the warning after Definition 2.3.1, $[3][3] = [9] = [3]$ illustrates the same.)

(c) **False:** (F4) fails. A zero $z = (z_1, z_2)$ would satisfy $(a,b)+z = (a,b)$ for all pairs; taking $(a,b) = (0,0)$ gives $(0,0)+z = (0+z_1,\ 0+z_2) = z$ by Theorem 1.2.3 (a), so $z = (0,0)$. But $(1,0) + (x,y) = (1+x,\ y) \neq (0,0)$ for every pair (x,y), since $1 + x = x + 1 \neq 0$ (Theorem 1.2.3 (b), Corollary 1.2.2). So $(1,0)$ has no negative. Passing to classes is what supplies negatives.

(d) **True.** Add $-a$:

$$b = 0 + b = ((-a) + a) + b = (-a) + (a + b) = (-a) + a = 0,$$

by (F1) and (F3), (F1) and (F4), (F2), the hypothesis, and (F1) and (F4) again.

2.3.3.3 C. Going deeper

Exercise. C1: A ring in disguise

On $\mathbb{Z}$ define $x \oplus y := x + y - 1$ and $x \odot y := x + y - xy$.

(a) Find the elements that play the parts of 0 and 1 in (F3) and (F7), and the negative of x in (F4).

(b) Prove (F9) for $\oplus$, $\odot$ directly.

(c) Let $\varphi(x) := 1 - x$. Show that $\varphi(x \oplus y) = \varphi(x) + \varphi(y)$, $\varphi(x \odot y) = \varphi(x)\varphi(y)$ and $\varphi(\varphi(x)) = x$, and use this to prove (F5) for $\odot$ without expanding.

Solution (Hint 1). In (a), solve $x \oplus e = x$ for e.

Solution (Hint 2). In (c), apply φ to both sides of (F5) and use that φ is injective.

Solution. All computations below are in the commutative ring $\mathbb{Z}$ (Theorem 2.3.1), with the rules of Proposition 2.3.2 (a)–(c). We also use $-(u+v) = (-u) + (-v)$: both sides are negatives of $u+v$, by (F1)–(F4), and negatives are unique (Proposition 2.3.2 (a)).

(a) The zero is 1: $x \oplus 1 = x + 1 - 1 = x$. The one is 0: $x \odot 0 = x + 0 - x \cdot 0 = x$. The negative of x is $2 - x$: $x \oplus (2 - x) = x + (2 - x) - 1 = 1$, the zero.

(b) On the left,

$$\begin{aligned} x \odot (y \oplus z) &= x + (y + z - 1) - x(y + z - 1) \\ &= 2x + y + z - 1 - xy - xz. \end{aligned}$$

On the right, $(x \odot y) \oplus (x \odot z) = (x + y - xy) + (x + z - xz) - 1$, which regroups to the same expression.

(c) The three identities:

$$\begin{aligned} 1-(x+y-1) &= (1-x)+(1-y),\\ 1-(x+y-xy) &= (1-x)(1-y),\\ 1-(1-x) &= x. \end{aligned}$$

(For the second, expand the right side: $1-y-x+xy$.) The third makes φ injective: if $\varphi(x)=\varphi(x')$, then $x=\varphi(\varphi(x))=\varphi(\varphi(x'))=x'$. Now

$$\begin{aligned} \varphi\big((x\odot y)\odot z\big) &= \big(\varphi(x)\varphi(y)\big)\varphi(z)\\ &= \varphi(x)\big(\varphi(y)\varphi(z)\big) = \varphi\big(x\odot(y\odot z)\big), \end{aligned}$$

by the second identity twice, (F5) in $\mathbb{Z}$, and the second identity twice more. Injectivity gives $(x\odot y)\odot z = x\odot(y\odot z)$.

The level below is now $\mathbb{Z}$ itself, and φ carries every law of $\mathbb{Z}$ across to $\oplus$, $\odot$.

Marking scheme. (10 points) - 3: (a) finds the zero 1, the one 0 and the negative $2-x$, each checked by a computation. - 3: (b) expands both sides to one expression, with the laws of $\mathbb{Z}$ used. - 1: (c) verifies the three identities for φ. - 1: (c) shows that φ is injective from $\varphi(\varphi(x))=x$, or otherwise uses $\varphi(\varphi(x))=x$ to recover arguments. - 2: (c) carries (F5) across φ and concludes, by injectivity or by applying φ once more ($\varphi\circ\varphi=\mathrm{id}_{\mathbb{Z}}$). Fatal errors (0 for the step): assuming the laws for $\oplus$, $\odot$ in (b); applying φ and concluding equality of the arguments without injectivity. A reader who proves (F5) for $\odot$ by direct expansion earns the 2 points of the last step only if the expansion is complete and correct, and loses none of the others.